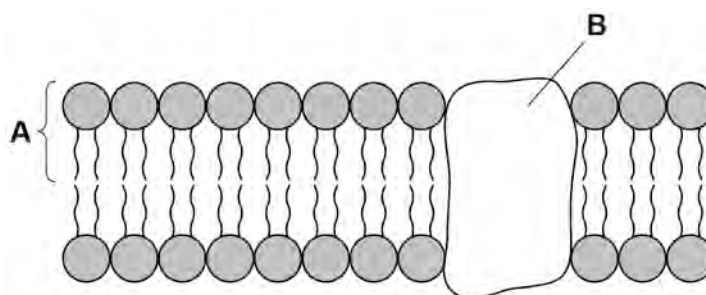


08.1

Figure 10 shows a diagram of a cell-surface membrane.

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Figure 10



Name **and** give the functions of the molecules labelled **A** and **B**.

[2 marks]

Molecule **A** name _____

Molecule **A** function _____

Molecule **B** name _____

Molecule **B** function _____



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A student investigated the effect of temperature on the permeability of cell-surface membranes in beetroot cells. Beetroot cells contain a red pigment that is not transported across intact cell-surface membranes.

The student:

- cut discs of beetroot tissue
- took five boiling tubes and placed two discs of beetroot into each boiling tube
- added water to the boiling tubes
- placed each boiling tube into a water bath at a different temperature for 30 minutes
- removed the discs and examined the water in each boiling tube.

Give **one** variable the student should have controlled in this investigation **and** explain why.

Do **not** refer to number of discs or length of time.

[2 marks]

Controlled variable _____

Explanation _____

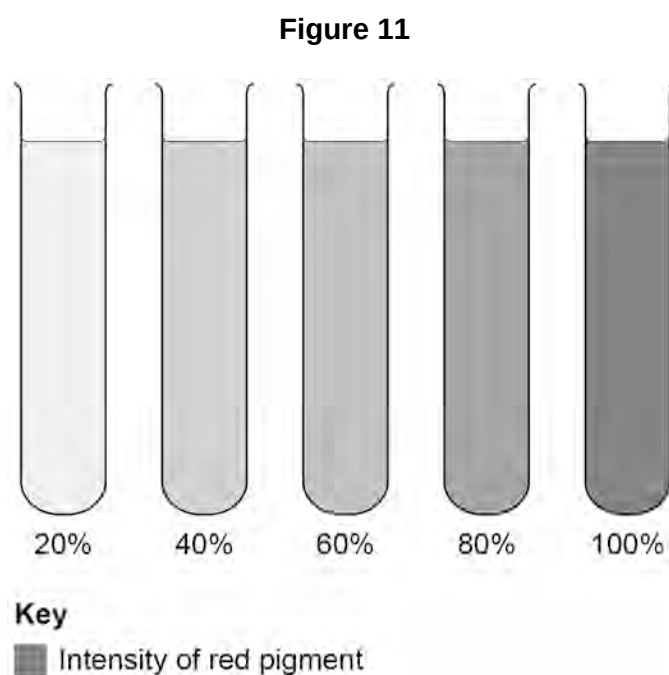
Question 8 continues on the next page

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Another student carried out the same investigation, ensuring that **all** variables were controlled.

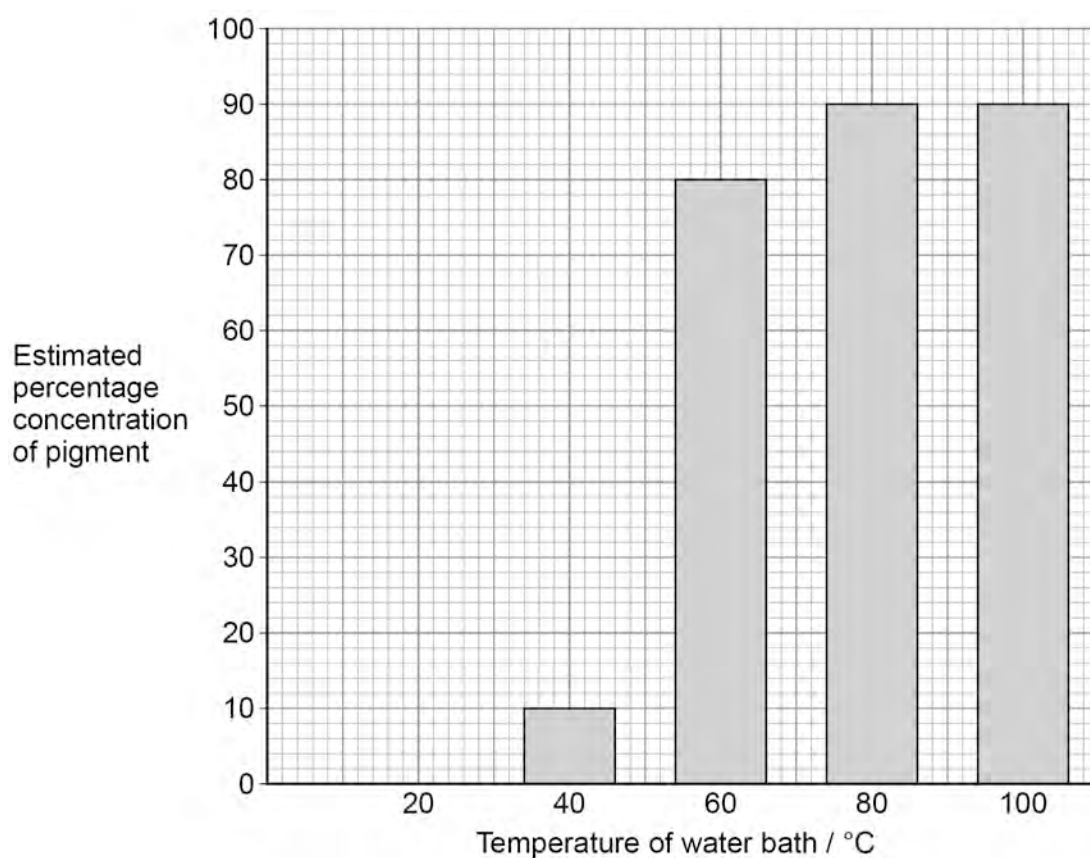
Figure 11 shows a dilution series the student prepared from 100% red beetroot pigment solution.



This student used the colours of the dilution series to estimate the percentage concentration of pigment in the boiling tubes at each temperature.

Figure 12 shows the student's results.



Figure 12**0 8 . 3**Describe and explain the general trend of the results in **Figure 12**.**[4 marks]**

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0 8 . 4

Describe how the student could use the dilution series (**Figure 11**, on page 22) and a colorimeter to determine an accurate measurement of the pigment concentration in the tubes at different temperatures.

[3 marks]

11

